TITLE: **Return Dynamics of ASELSAN under Global and Sovereign Risk Factors: co-movement and one-day-ahead forecasts**

SUBJECT: Aim, method and results of the analyses in version 2 of the repository `Return-Dynamics-of-ASELSAN-under-Global-and-Sovereign-Risk-Factors`

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1. Aim and scope

Version 1 of this project studies the daily return of ASELSAN (ticker ASELS, Borsa Istanbul) with an ARX model: its own past returns and eight factors (BIST100, USD/TRY, Bitcoin, Brent oil, VIX, the dollar index DXY, the US 10-year yield and Turkey's 5-year sovereign CDS spread), each with lags $0, 1, 2$, estimated by OLS with Newey–West standard errors. It then compares the ARX model with simple benchmarks and four machine-learning models over 2025 and early 2026, and concludes that ASELSAN's return has "modest but robust" out-of-sample predictability. Version 2 keeps the variables, the transformations, the lag orders, the HAC lag, the sample split and the model hyperparameters of version 1, and separates two questions that version 1 treated as one:

1. **How does ASELSAN move with the factors on the same day?** This is what an ARX model with lag-0 factors measures. Analysis 2.
2. **Can ASELSAN's return be forecast one day ahead?** A forecast for day t may only use what is known at the close of day $t-1$. Analysis 3.

Analysis 1 freezes the data, describes how series traded on different calendars are aligned, and reproduces version 1.

2. Data and notation

Series. Daily closes from Yahoo Finance for ASELS.IS, XU100.IS (BIST100), TRY=X (USD/TRY), BTC-USD, BZ=F (Brent), ^VIX, DX-Y.NYB (DXY) and ^TNX (US 10-year yield, in per cent), downloaded once for the window of version 1 (2020-01-01 to 2026-01-12) and stored in `data/yahoo_close.csv`; and the Turkey 5-year sovereign CDS spread in basis points, the file of version 1, which ends on 2026-01-08. The source of the CDS file was not recorded in version 1; it is most likely Investing.com.

Alignment. Let $\mathcal{T}$ be the set of ASELSAN trading days and $\mathcal{T}_s$ the dates on which series s has a close P^s_u. On day $t \in \mathcal{T}$ series s enters with its latest close on or before t:

$$X^s_t = P^s_{\tau_s(t)}, \qquad \tau_s(t) = \max\{u \in \mathcal{T}_s : u \le t\}. \tag{1}$$

Nothing is filled backwards, and the panel ends on the last ASELSAN day on which every series has its own data, 2026-01-08; it has 1,507 days. The age $t - \tau_s(t)$ is positive when the market of s was closed on t (Table 1 of `figures/`): on 52 days (3.5%) for VIX, 49 for Brent, 50 for DXY, 52 for the US yield, 8 for CDS, 5 for BIST100 and 3 for USD/TRY.

Transformations (as in version 1). For the price-like series $s \in \{\text{ASELS}, \text{BIST100}, \text{USDTRY}, \text{BTC}, \text{BRENT}\}$ the return in per cent is $r^s_t = 100\,(\ln X^s_t - \ln X^s_{t'})$, and for $s \in \{\text{VIX}, \text{DXY}, \text{US10Y}, \text{CDS}\}$ the change is $\Delta X^s_t = X^s_t - X^s_{t'}$, where t' is the previous ASELSAN day. Write $y_t = r^{\text{ASELS}}_t$ and $Z_t = (Z_{1,t}, \ldots, Z_{8,t})'$ for the eight transformed factors. The returns run from 2020-01-03 to 2026-01-08 (1,506 days).

Timing. Borsa Istanbul closes before the US markets, Brent futures and the daily Bitcoin bar do. A factor value dated t is therefore known after ASELSAN's close of day t, sometimes several hours after it.

3. Models and tests

ARX model. For lag orders p and q and a first factor lag j_0,

$$y_t = \alpha + \sum_{i=1}^{p} \phi_i\, y_{t-i} + \sum_{k=1}^{8} \sum_{j=j_0}^{q} \beta_{k,j}\, Z_{k,t-j} + \varepsilon_t = x_t'\theta + \varepsilon_t. \tag{2}$$

$j_0 = 0$ is the model of version 1: it contains Z_t, which is not known before day t ends, so it describes *co-movement*. $j_0 = 1$ is the *forecasting* model: every regressor is known at the close of day $t-1$. Version 1 uses $p = q = 2$, and so does version 2 unless stated otherwise.

Estimation and standard errors. $\hat\theta = (X'X)^{-1}X'y$, with the Newey–West (1987) covariance with Bartlett weights and $L = 5$ lags, as in version 1:

$$\begin{aligned} \hat V &= (X'X)^{-1}\,\hat S\,(X'X)^{-1}, \\ \hat S &= \sum_t \hat u_t^2 x_t x_t' + \sum_{\ell=1}^{L} \Big(1 - \frac{\ell}{L+1}\Big) \sum_t \hat u_t \hat u_{t-\ell}\,\big(x_t x_{t-\ell}' + x_{t-\ell} x_t'\big). \end{aligned} \tag{3}$$

Channel tests. Version 1 states, for each factor k, the hypothesis $H_0 : \beta_{k,j} = 0$ for all j, but tests single coefficients and, for VIX and CDS, the sum of their coefficients. Version 2 tests it jointly. With R the matrix that selects the r coefficients of factor k,

$$W = (R\hat\theta)'\,(R\hat V R')^{-1}\,(R\hat\theta) \;\xrightarrow{d}\; \chi^2_r \quad \text{under } H_0. \tag{4}$$

The cumulative effect $s_k = \sum_j \beta_{k,j}$ (the response of y_t to a lasting unit change of Z_k) is tested with the same formula and $R = (0, \dots, 1, \dots, 1, \dots, 0)$, $r = 1$. To compare factors measured in different units, s_k is also reported per standard deviation of Z_k: $\hat{s}_k \hat{\sigma}_{Z_k}$, with the interval $(\hat{s}_k \pm 1.96 \sqrt{R \hat{V} R'})\, \hat{\sigma}_{Z_k}$.

Recursive forecasts. The test sample is $\mathcal{T}_{\text{test}}$ = 2025-01-02 to 2026-01-08 (256 days). For every $t \in \mathcal{T}_{\text{test}}$ every model is fitted on all rows dated before t (expanding window, refitted every day) and applied to x_t. The models are: zero, $\hat{y}_t = 0$; the prevailing mean, $\bar{y}_t = \frac{1}{n_t} \sum_{u<t} y_u$ (Welch and Goyal, 2008); the random walk, $\hat{y}_t = y_{t-1}$; AR(p), the model (2) without factors; the ARX model; and Ridge, Lasso, random forest and XGBoost given exactly the inputs of the ARX model. The hyperparameters of the tree models are those of version 1; the Ridge and Lasso penalties are chosen from the grid of version 1, $10^{-4}, \dots, 10^{4}$ (50 values), by five-fold cross-validation in which each fold is validated only on data after its training data.

Forecast evaluation. With $e_t = \hat{y}_t - y_t$:

- the out-of-sample R^2 against the prevailing mean (Campbell and Thompson, 2008), $R^2_{\text{OS}} = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2$; it is positive when the model has the smaller squared error;
- the Diebold–Mariano (1995) test of equal squared-error loss with the correction of Harvey, Leybourne and Newbold (1997). For one-step forecasts and $d_t = e_{1,t}^2 - e_{2,t}^2$,

$$\begin{aligned} &\text{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}_d^2 / T}} \sqrt{\frac{T-1}{T}}, \qquad \hat{\sigma}_d^2 = \frac{1}{T} \sum_t (d_t - \bar{d})^2, \\ &\text{DM} \sim t_{T-1} \text{ (two-sided);} \end{aligned} \tag{5}$$

- the Clark–West (2007) test for nested models. If the small model (forecast $\hat{y}_{1,t}$) is nested in the large one ($\hat{y}_{2,t}$), the large model's squared error is biased upwards under the null by the noise of its extra estimated coefficients. The test adjusts for this,

$$f_t = (y_t - \hat{y}_{1,t})^2 - \left[(y_t - \hat{y}_{2,t})^2 - (\hat{y}_{1,t} - \hat{y}_{2,t})^2 \right], \qquad \text{CW} = \frac{\bar{f}}{s_f / \sqrt{T}}, \tag{6}$$

  and rejects equal accuracy in favour of the large model when CW exceeds the one-sided normal critical value;
- directional accuracy $\hat{P}$, the share of days with $\operatorname{sign} \hat{y}_t = \operatorname{sign} y_t$ (days with $y_t = 0$ or $\hat{y}_t = 0$ excluded), and the Pesaran–Timmermann (1992) test of independence between predicted and realized signs. With $\hat{P}_y$, $\hat{P}_f$ the shares of positive returns and positive forecasts, $P_* = \hat{P}_y \hat{P}_f + (1 - \hat{P}_y)(1 - \hat{P}_f)$ is the accuracy expected by chance,

and

$$\mathrm{PT} = \frac{\hat{P} - P_*}{\sqrt{\widehat{\mathrm{Var}}(\hat{P}) - \widehat{\mathrm{Var}}(P_*)}} \sim N(0,1) \text{ (one-sided).} \tag{7}$$

A forecast that never changes sign has no PT statistic: its accuracy is the share of days that went its way.

4. Analysis 1: The data and version 1 reproduced (Figure 1)

Why. Version 1 downloaded the Yahoo series at every run, up to the day of the run, so its numbers could not be reproduced, and its last row was taken during the trading day.

Method. The series are downloaded once and stored with the versions of the software used. The code of version 1 is run unchanged on the stored data (`aselsan/legacy.py`). Its last row (12 January 2026) is compared with the final closes of that day.

How to read the figure. (a) $\sum_{u \le t} r_u$ for ASELSAN and BIST100. (b) The slope of r^{ASELS} on r^{BIST} in the previous 120 days, as in version 1.

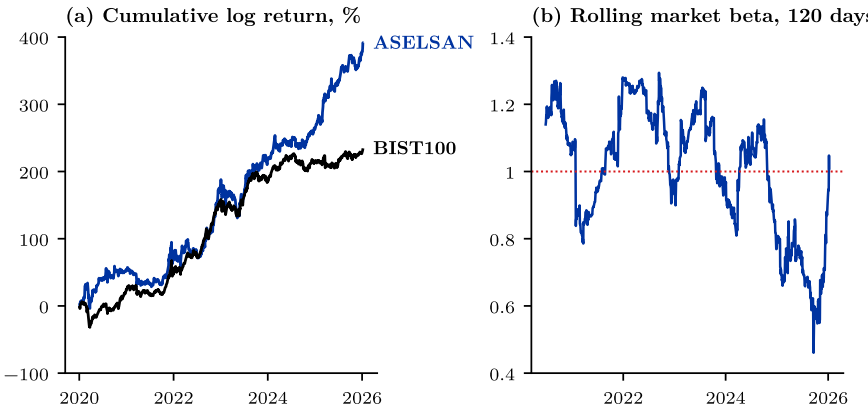


Figure 1. ASELSAN and BIST100, 2020-01-03 to 2026-01-08.

Finding. Version 1's code on the frozen data gives its printed results to within 0.01 for R^2, the market coefficient and the RMSEs of all models except XGBoost (Table 1); the frozen data have one ASELSAN day fewer than the data of version 1. XGBoost differs most, by 0.07. In the last row of version 1 Brent was 63.40 against a final close of 63.87, and VIX 15.17 against 15.12; the CDS file had no value for that day, so version 1 carried the value of 8 January forward, which made the last two CDS changes zero. Over the panel ASELSAN's cumulative log return was 391% against 234% for BIST100; its daily standard deviation was 2.78 per cent against 1.77, the correlation 0.66. The 120-day market beta moved between 0.46 and 1.29.

Table 1. Version 1: printed results and the same code run on the frozen data.

Quantity	Printed by version 1	Version 1's code, frozen data
Observations in the ARX model	1,507	1,506
R^2 of the ARX model	0.4477	0.4478
Coefficient of r_t^{BIST}	1.0494	1.0495
p-value of r_{t-1}^{USD}	0.0334	0.0343
RMSE, ARX	2.4903	2.4945
RMSE, zero forecast	2.7440	2.7493
RMSE, AR	2.7099	2.7158
RMSE, Lasso	2.4746	2.4794
RMSE, random forest	2.4982	2.5035
RMSE, XGBoost	2.6994	2.6252
Directional accuracy of ARX (%)	64.86	65.12
Test days	259	258
DM p-value, ARX against zero	0.0424	0.0419

Interpretation. The results of version 1 are reproducible to the stated precision. Version 2 uses the frozen data and ends the panel on the last CDS date, so that no series is carried past its own end.

5. Analysis 2: Co-movement on the same day (Figure 2, Table 2)

Why. The ARX model of version 1 ($j_0 = 0$) describes how ASELSAN moves with the factors on the same day. Version 1 read it channel by channel from single coefficients.

Method. Model (2) with $j_0 = 0$, $p = q = 2$, HAC with 5 lags; joint Wald test and cumulative effect for each channel; the same model with $j_0 = 1$; in-sample R^2 of both for $p = q \in \{1, 2, 3, 5\}$; and the robustness grid of version 1 (VIX and CDS in levels or in changes, $p = q \in \{1, 3, 5\}$).

How to read. Table 2: p-values of the joint test of all coefficients of a channel and of its cumulative effect; the effect of a one-standard-deviation change in percentage points of ASELSAN return, with its 95% interval; and the joint test when the lag-0 term is removed. Figure 2(a): R^2 with and without the same-day factors. (b): the effects of the table.

Finding. The model of version 1 is reproduced: $R^2 = 0.448$, market coefficient 1.049 (s.e. 0.041). Almost all of the fit comes from the same day: without the lag-0 terms R^2 falls from 0.448 to 0.021 (adjusted 0.010), and with five lags it is 0.051 (Figure 2a). Tested jointly, the exchange-rate channel is significant on the same day ($p = 0.034$) and is the only lagged channel that is ($p = 0.0039$); the other lagged channels have $p \geq 0.13$. VIX and Brent are not significant as channels ($p = 0.138$ and 0.094), but their cumulative effects are ($p = 0.030$ and 0.015): a one-standard-deviation rise of VIX (2.23 points) goes with 0.25 [0.03,

Table 2. Channels in the ARX model, $p = q = 2$, $N = 1,504$, Newey–West with 5 lags. Effects in percentage points of the ASELSAN return per standard deviation of the factor, lags 0–2.

	Same-day model ($j_0 = 0$)			Lagged ($j_0 = 1$)
Channel	joint p	cumulative p	effect per s.d. [95% CI]	joint p
market (BIST100)	<0.0001	<0.0001	–	0.6293
exchange rate (USD/TRY)	0.0335	0.1382	0.14 [−0.04, 0.32]	0.0039
global risk (VIX)	0.1376	0.0296	0.25 [0.03, 0.48]	0.6661
sovereign risk (CDS)	0.3079	0.5082	−0.06 [−0.23, 0.12]	0.3294
dollar (DXY)	0.4630	0.4362	−0.09 [−0.31, 0.14]	0.1591
US 10-year yield	0.4279	0.1574	−0.15 [−0.36, 0.06]	0.6184
Bitcoin	0.7087	0.9852	0.00 [−0.18, 0.18]	0.8296
Brent oil	0.0937	0.0153	0.23 [0.04, 0.41]	0.1328

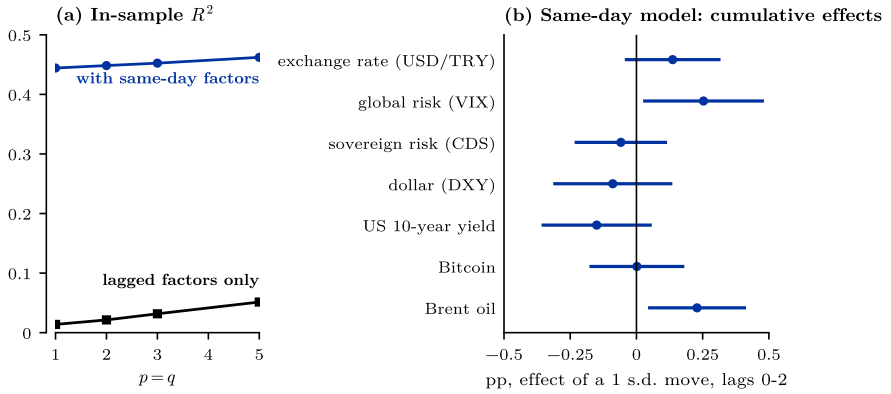


Figure 2. (a) In-sample R^2 of model (2) with $j_0 = 0$ and $j_0 = 1$. (b) Cumulative effect of a one-standard-deviation change of each factor (lags 0–2) with 95% HAC interval.

0.48] percentage points higher ASELSAN return over three days, and of Brent (2.69%) with 0.23 [0.04, 0.41]. The CDS channel is not significant in any form in the baseline: joint $p = 0.308$, cumulative $p = 0.508$, effect −0.06 [−0.23, 0.12]. In the robustness grid the market coefficient stays between 1.040 and 1.053; the cumulative effect of CDS changes has p between 0.067 and 0.920; the sum of the CDS level coefficients has p between 0.009 and 0.041.

Interpretation. ASELSAN is a market stock with a beta near one; beyond the market, its same-day return moves with the exchange rate, and its three-day cumulative response is positive to VIX and to oil. A joint test asks whether any coefficient of the channel differs from zero; the cumulative test asks about one direction, their sum, and has more

power when the coefficients share a sign. With a lower bound of 0.03, the VIX effect is imprecisely estimated. The negative CDS result that version 1 reported does not survive in changes. In levels the sum of the coefficients is the coefficient on a very persistent regressor, for which the normal approximation behind these p-values is not reliable, so that result is not taken as evidence. None of this is a forecast: the same-day values are not known in advance, and the US and oil closes of day t are set after Borsa Istanbul closes.

6. Analysis 3: One-day-ahead forecasts (Figure 3, Table 3)

Why. Version 1's "out-of-sample forecasts" for day t used the factors of day t, and its machine-learning models were given every column of its data table, including the same-day prices and returns of the factors and the closing price of ASELSAN on day t. They measure how well the return can be explained once the day is over. Version 1 reported RMSE 2.49 and directional accuracy 64.9% for its ARX model, and a Diebold–Mariano $p = 0.042$ against the zero forecast.

Method. Recursive forecasts for the 256 test days (first training sample: 1,248 days). In the forecasting design ($j_0 = 1$, $p = q = 2$) every input is known at the close of day $t - 1$. For comparison, the same-day design of version 1 and a model with the same-day BIST100 return alone. The benchmark is the prevailing mean, the standard benchmark in this literature (Welch and Goyal, 2008; Rapach and Zhou, 2013); Clark–West is used for the nested linear models.

How to read. Table 3: RMSE in percentage points; R^2_{OS} against the prevailing mean; directional accuracy (%) and the PT p-value; DM p-value against the prevailing mean; CW p-value against the prevailing mean. Figure 3: $\sum_{u \le t}[(y_u - \bar{y}_u)^2 - (y_u - \hat{y}_u)^2]$; a rising line means the model is gaining on the prevailing mean.

Table 3. One-day-ahead forecasts, 2025-01-02 to 2026-01-08 (256 days), recursive and refitted daily.

Model	RMSE	R^2_{OS}	Dir. acc.	PT p	DM p	CW p
Zero	2.730	−0.024	–	–	0.0299	–
Prevailing mean	2.698	0.000	55.3	–	–	–
Random walk	3.833	−1.019	47.6	0.8278	<0.0001	–
AR	2.703	−0.004	55.3	0.3003	0.6890	0.5108
ARX	2.757	−0.044	51.4	0.6428	0.0264	0.8979
Ridge	2.705	−0.005	55.3	0.3653	0.4154	–
Lasso	2.704	−0.004	55.3	0.4398	0.2247	–
Random forest	2.715	−0.013	55.7	0.2144	0.4535	–
XGBoost	2.962	−0.205	49.4	0.6705	0.0010	–
Same-day factors (design of version 1, not a forecast)						
ARX, same-day factors	2.482	0.154	65.1	<0.0001	0.0854	–
BIST100 same day only	2.467	0.164	64.7	<0.0001	0.0745	–

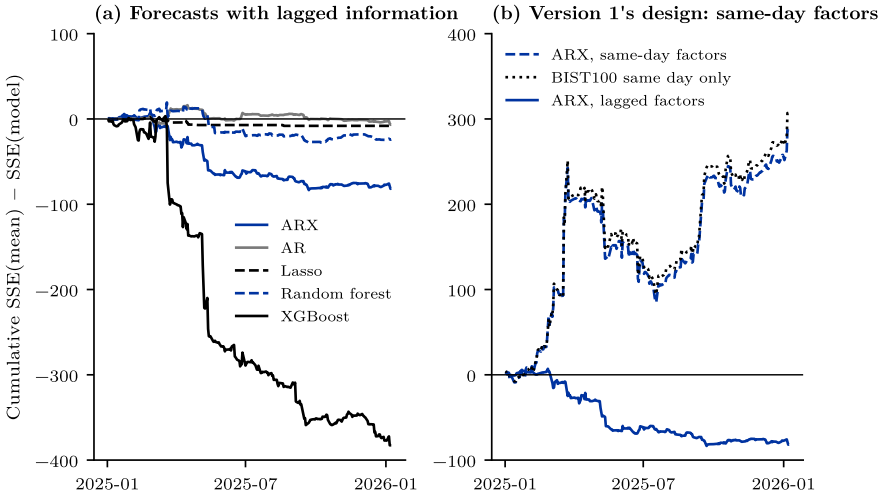


Figure 3. Cumulative difference of squared errors, prevailing mean minus model. (a) Forecasts with information up to $t-1$. (b) The same-day design of version 1 against the lagged ARX model.

Finding. With information up to day $t-1$ no model beats the prevailing mean. The ARX model has $R^2_{\text{OS}} = -0.044$ and is significantly worse than the mean (DM $p = 0.0264$) and than the AR model (DM $p = 0.032$); Clark–West finds no gain from the factors ($p = 0.898$ against the mean, 0.935 against AR). Ridge, Lasso and AR are close to the mean (R^2_{OS} -0.005, -0.004, -0.004): they forecast a rise on at least 91% of the days, and their directional accuracy, 55.3%, equals the share of rising days among the 255 non-zero returns, 55.3%. The random forest has -0.013; XGBoost, -0.205, is significantly worse than the mean (DM $p = 0.0010$). No directional accuracy differs from chance (PT $p \geq 0.21$). More lags make the ARX model worse: R^2_{OS} is -0.024, -0.044, -0.063 and -0.076 for $p = q = 1, 2, 3, 5$. With the same-day factors of version 1 the picture reverses: $R^2_{\text{OS}} = 0.154$ and directional accuracy 65.1% (PT p <0.0001), close to version 1's results. The same-day BIST100 return alone does as well ($R^2_{\text{OS}} = 0.164$, 64.7%).

Interpretation. The accuracy reported by version 1 comes from the same-day market return, which is not known when the forecast would have to be made. Once the information set is restricted to day $t-1$, the factors carry no forecasting power for ASELSAN's daily return in the test period; the models whose forecasts stay close to the mean (AR, Ridge, Lasso) lose least, and XGBoost loses most. This agrees with the general finding that daily stock returns are hard to forecast beyond their historical mean (Welch and Goyal, 2008; Rapach and Zhou, 2013). Two points limit the strength of this conclusion: the test sample is one year, and returns are heavy-tailed, so a few days carry a large share of the squared errors; Figure 3(b) shows the jumps of single days. Even the same-day R^2_{OS} of 0.154 has a DM p-value of only 0.0854.

7. Conclusions and limitations

- ASELSAN's daily return moves with the market (beta 1.049) and, beyond it, with the exchange rate; its cumulative same-day response to VIX and Brent is positive; sovereign CDS changes show no significant relation in the baseline.
- These are same-day relations. With information up to the previous close, neither the ARX model nor Ridge, Lasso, random forest or XGBoost forecasts the next day's return better than the prevailing mean in 2025–2026.
- Limitations: one test year; a single stock; the factor closes of the US and oil markets are set after Borsa Istanbul closes, so even the same-day relations mix information from different hours; the CDS series comes from a file whose source was not recorded; Yahoo's adjusted prices can be revised, which is why the downloaded data are stored with the repository.

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All numbers are generated from `figures/*.csv` by `make_paper.py`.